EE 66 Fall 2026 — Signals, Dynamics, and Information — Discussion 8B

1. Short Proofs

Let's define $\mathbf{S} = \mathbf{A}^\top \mathbf{A}$, where $\mathbf{A} \in \mathbb{R}^{m \times n}$ is an arbitrary real $m \times n$ matrix. $\mathbf{V} = \begin{bmatrix} \mathbf{v}_1 \cdots \mathbf{v}_n \end{bmatrix}$ is the matrix of normalized eigenvectors of $\mathbf{S}$ with eigenvalues $\lambda_1, \cdots, \lambda_n$.

(a) Let $\mathbf{v_1}, \cdots, \mathbf{v_r}$ be those normalized eigenvectors that correspond to non-zero eigenvalues (i.e. $\|\mathbf{v}_i\| = 1$ and $\lambda_i \neq 0$ for $i = 1, \cdots, r$). Show that $\|\mathbf{A}\mathbf{v}_i\|^2 = \lambda_i$.

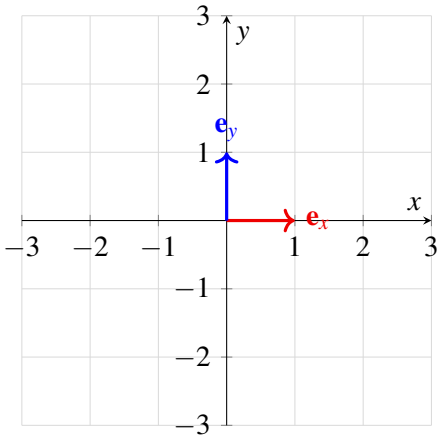
(b) Following the assumptions in part (a), show that $\mathbf{A}\mathbf{v}_i$ is orthogonal to $\mathbf{A}\mathbf{v}_j$.
Hint: It may help to recall the Spectral Theorem.

(c) Show that if $\mathbf{V} \in \mathbb{R}^{n \times n}$ is an orthonormal square matrix, then $||\mathbf{x}||^2 = ||\mathbf{V}\mathbf{x}||^2$ for all $\mathbf{x} \in \mathbb{R}^n$.
Hint: Write the norm as an inner product instead of trying to do this elementwise.

UCB EE 66, Fall 2026, Discussion 8B, All Rights Reserved. This may not be publicly shared without explicit permission.

2. Geometric Interpretation of the SVD

(a) When we plot the transformation given by a specific matrix, we think about how the matrix transforms the standard basis vectors. In 2D, let $\mathbf{e}_x = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\mathbf{e}_y = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. The vectors $\mathbf{e}_x$ and $\mathbf{e}_y$ are shown below



Consider the following matrix

$$\mathbf{A} = \begin{bmatrix} -1 & 0 \\ 0 & 2 \end{bmatrix}$$

How would A transform $\mathbf{e}_x$ and $\mathbf{e}_y$? Plot the result.

(b) Let's take a look at the 2×2 rotation matrix. Recall that this matrix will rotate any vector counter-clockwise by θ.

$$\mathbf{R} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \tag{1}$$

Show that this matrix is orthonormal.

(c) Now let's consider how this transformation looks in the lens of the SVD. You are given the following matrix $\mathbf{A}$:

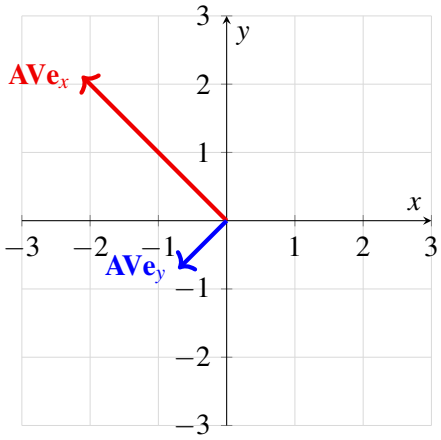
$$\mathbf{A} = \begin{bmatrix} -1 & -2 \\ 2 & 1 \end{bmatrix} \tag{2}$$

Recall that the SVD of this matrix is given by $\mathbf{A} = \mathbf{U}\Sigma\mathbf{V}^\top$. Assume you are told that

$$\mathbf{V} = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \tag{3}$$

We will try to deduce $\mathbf{U}$ and Σ graphically, and the confirm our results numerically. **Plot the transformation given by $\mathbf{V}$ by showing how it affects $\mathbf{e}_x$ and $\mathbf{e}_y$ via left multiplication.** *(HINT: Try writing $\mathbf{V}$ as a rotation matrix with a specific θ.)*

(d) Suppose you are told that the transformation of $\mathbf{AV}$ on $\mathbf{e}_x$ and $\mathbf{e}_y$ looks like



Write this transformation $\mathbf{AV}$ in terms of $\mathbf{U}$ and Σ. Recall that the $\mathbf{U}$ matrix is an orthonormal matrix so it will correspond to any rotations or reflections, and the Σ matrix is a diagonal matrix and will perform any scaling operations. **Based on this fact and the plot of the transformation above, write down a guess for what $\mathbf{U}$ and Σ might be.**

(e) **Based on the given V matrix, compute the SVD.** Does your answer match your hypothesis from the previous part?

(f) **Using your answer for U and Σ from the previous part, plot the transformation of Σ on $\mathbf{e}_x$ and $\mathbf{e}_y$. From here, plot the transformation of $\mathbf{U}\Sigma$ on $\mathbf{e}_x$ and $\mathbf{e}_y$.** Does the final plot resemble the transformation shown by $\mathbf{AV}$?

UCB EE 66, Fall 2026, Discussion 8B, All Rights Reserved. This may not be publicly shared without explicit permission.